

KrishiDrishti

Satellite-Based Crop Mapping, Moisture-Stress Detection & Irrigation Advisory

A full-stack AI + remote-sensing system: draw a box anywhere in northern India and get a per-field crop map, growth-stage-aware moisture stress, 8-day crop-water deficit and an irrigation advisory - in about 2-3 minutes. Built for ISRO Problem Statement 6 and extended into a deployable web service.

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1 Project Overview

Irrigation timing is one of the highest-leverage decisions in Indian agriculture, yet field-level guidance rarely exists: crop maps are coarse or stale, stress detection ignores the crop's growth stage, and water-deficit numbers are seldom translated into an actionable "irrigate now / wait" call. This project builds that missing chain end-to-end, from raw satellite pixels to a per-field advisory, and wraps it in a web application:

- **Crop-type classification** - an 8-class per-field crop map (Wheat, Mustard, Lentil, No-crop/Fallow, Sugarcane, Maize, Rice, Other) from a full season of Sentinel-1 SAR + Sentinel-2 optical time series.
- **Moisture-stress detection** - growth-stage-aware stress levels per field per 10-day dekad, fusing a physics-based water balance with observed spectral/SAR stress indicators.
- **8-day water deficit + irrigation advisory** - FAO-56 deficit in mm and m³ per 8-day period, translated into a 5-level advisory (Out of season / No stress / Watch / Irrigation advised / Severe).
- **A web service** - users draw an area of interest (AOI) on a map; the backend fetches that season's imagery from Google Earth Engine, runs the 11-stage pipeline, and returns an interactive dashboard with color-coded field-level maps and downloadable GeoTIFF/CSV products.

Headline numbers

Metric	Value
Crop classification (held-out test)	OA 0.78 · macro-F1 0.55 · Cohen's kappa 0.68
Honest generalization (5-fold chip-level CV)	OA ~0.73
Sowing-date detection (NDVI green-up)	87% of fields observed directly
LSTM stress emulator vs FAO-56 Ks (held-out chips)	R ² 0.96 · MAE 0.036 · stress-flag P 0.94 / R 0.95
End-to-end AOI job (5x5 km, incl. GEE fetch)	~140-160 s
Demonstrator scale (Uttar Pradesh pilot)	537 chips · 2,992 fields · 18 dekads · 1.3 Mm ³ season deficit over 793 ha

Every detection step is **causal** (no future dekads are used), so the same pipeline can run mid-season on a live crop, not just retrospectively.

2 Dataset Summary

Source	Content	Role
AgriFieldNet India 2021-22	5,771 labeled fields, 13 raw crop classes, 4 pilot regions; avg field ~0.26 ha (~26 px at 10 m)	Ground truth for training / validation

Sentinel-2 L2A (via GEE)	11 monthly composites Jun 2021 - Apr 2022; 9 bands; s2cloudless + SCL masking, median composite	Optical signal: NDVI, NDMI, red-edge, EVI, gNDWI + raw bands
Sentinel-1 IW GRD	11 monthly VV/VH backscatter composites (dB)	All-weather structure / moisture; region-normalized
Dekadal SEASON composites	18 ten-day composites (B04/B05/B08), Nov-Apr	Fine phenology: sowing green-up, dekadal NDVI/NDRE
NASA POWER daily meteorology	T, RH, wind, solar, rain on a 0.5 deg grid	FAO-56 Penman-Monteith ET_0 for the water balance

Training corpus: 1,151 chips of 256x256 px at 10 m (T = 11 months) across Uttar Pradesh (537), Rajasthan (238), Odisha (200) and Bihar (176). Splits are **chip-level stratified 80/10/10** (stratum = region x rarest-present crop) so spatially-correlated pixels never leak across train and test. Class consolidation: 13 raw crops to 8 classes; minor crops (green pea, garlic, gram, coriander, ...) fold into "Other".

3 Workflow

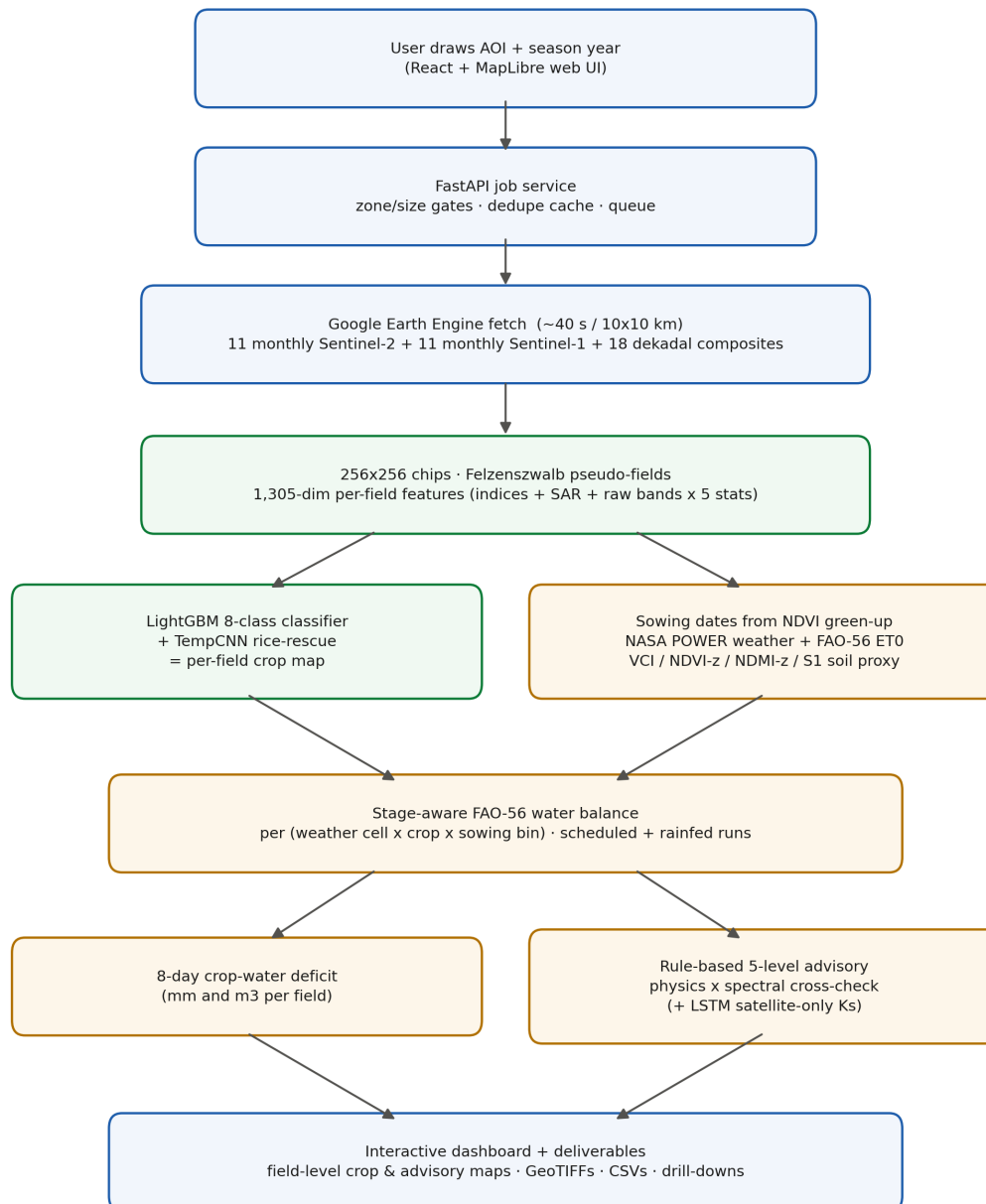


Figure 1 - End-to-end workflow: from a user-drawn AOI through GEE compositing, classification and the FAO-56 water balance to the advisory dashboard.

4 Tech Stack

Front-end

- **React 18 + Vite** single-page app; **MapLibre GL JS** for the satellite basemap, drag-to-draw AOI, supported-zone mask and pilot-area badges (VALIDATED vs EXPERIMENTAL).
- Live 11-stage progress (2 s polling), results panel with KPIs, crop-mix chart, deficit sparkline and file downloads.

Back-end

- **FastAPI + Uvicorn** job API; also serves the built frontend, the dashboards and all output files from a single process.

- **Google Earth Engine Python API:** ~40 parallel computePixels requests fetch a full season for a 10x10 km AOI in ~40 s.
- Production hygiene: job dedupe by (bbox, year) hash with instant cache hits, one active job per client, rate limiting, a global concurrency semaphore, resumable stages, and traversal-guarded file serving.

ML / geospatial

- **LightGBM** (classifier) · **PyTorch** (TempCNN, LSTM) · **scikit-learn / scikit-image** (metrics, Felzenszwalb segmentation) · **rasterio** (GeoTIFF I/O) · **NumPy, matplotlib, Pillow**.
- Self-contained HTML dashboard (inline SVG charts, no framework), served by the API or opened as a local file.

5 Model & System Architecture

5.1 Crop classifier - LightGBM on engineered time-series features

- **Features (1,305-dim per field):** 261 per-pixel features x 5 field statistics (mean/std/median/min/max). Blocks: NDVI/NDMI monthly series, cloud-mask fractions, region-normalized VV/VH/ratio SAR series, red-edge indices (NDRE, Clre, RE-NDVI), EVI + gNDWI, **99 raw S2 band-month values** (the single biggest accuracy lever found), kharif SAR-flooding cues and static context.
- **Model:** LightGBM - 500 trees, 31 leaves, lr 0.05, class-balanced weights.
- **Rice-rescue specialist:** a 13-class **TempCNN** (temporal CNN over the 7-channel monthly sequence + 8 static features) re-examines fields the LGB called “Fallow” and flips them to Rice when $P(\text{rice}) > 0.70$ - recovering a rare class the tabular model under-predicts.
- **Validation discipline:** chip-level splits (no spatial leakage); the reported CV OA ~0.73 is the honest generalization number. Error analysis shows the ceiling is dominated by Wheat-Mustard spectral overlap at 10 m on 0.26 ha smallholder fields - a data limitation, not a tuning one (confirmed across RF/XGBoost/deep baselines, texture, phenology and ensembling experiments).

5.2 Sowing-date detection - per-field phenology

NDVI green-up crossing (threshold 0.30) on the dekadal series, **kharif-trough aware** (a field already green in November must first dip and re-cross, so standing kharif crop is not mistaken for rabi sowing), minus a crop-specific emergence lag. 87% of fields are observed directly; the rest fall back to crop-median or literature calendars. The observed wheat sowing spread (p10-p90: Oct 20 to Dec 13) shifts a field's irrigation requirement by up to ~3x, which is why per-field timing matters.

5.3 Water balance & advisory - physics first, ML where physics can't reach

- **FAO-56 dual-run water balance** per (weather cell x crop x sowing bin): stage-dependent Kc curves, root-zone bucket, depletion vs the readily-available-water (RAW) trigger. A *scheduled* run yields the advised irrigation calendar; a *rainfed* run yields the stress coefficient Ks and the 8-day deficit (ETc - ETa).
- **Advisory fusion - fixed, auditable rules (deliberately not a learned model):** “Irrigation advised” fires when the bucket hits the RAW trigger in a water-sensitive stage, or when moderate spectral stress (VCI / NDVI-z / NDMI-z) coincides with a dry Sentinel-1 soil-moisture proxy or $K_s < 0.85$; “Severe” requires $K_s < 0.85$ **and** strong spectral stress in a sensitive stage; “Watch” at 80% of trigger depletion or mild spectral stress. Physics and observation must corroborate each other before a farmer is told to spend water.
- **LSTM Ks emulator (satellite-only fallback):** a 2-layer causal LSTM (hidden 64, 19 spectral/SAR input features, **no weather inputs**) distilled against the FAO-56 rainfed Ks. Held-out chips: MAE 0.036, R^2 0.96, stress-flag precision 0.94 / recall 0.95. Its role is stress estimation where meteorological data is missing or late; the dashboard automatically flags (and the advisory ignores it) when its agreement with the water balance is poor on out-of-distribution areas or seasons.

5.4 Serving arbitrary AOIs

The training pipeline is parameterized end-to-end via environment variables, so identical code serves any bbox: GEE composites use the **same compositing recipe as the training tiles** (verified band-for-band), pseudo-fields come from Felzenszwalb segmentation on peak-NDVI (cadastral boundaries don't exist for arbitrary AOIs), and a northern-India gate (68-89 deg E, 19-32.5 deg N) blocks requests where the model has zero training support. The four pilot areas are badged VALIDATED; everywhere else is explicitly EXPERIMENTAL.

6 Results & Deliverables

Deliverable	Form
Crop-type map	Field-level 10 m mosaic PNG + per-chip GeoTIFFs with colormap
Stage-wise stress / advisory maps	18 per-dekad translucent overlays + GeoTIFFs
8-day water deficit	Per-field mm & m ³ series, CSVs, 23-band deficit GeoTIFFs
Irrigation advisory	5-level per-field per-dekad, CSV + dashboard ribbon
Dashboard	Self-contained HTML: KPIs, season timeline, demand-vs-rainfall, sowing histograms, per-field drill-downs

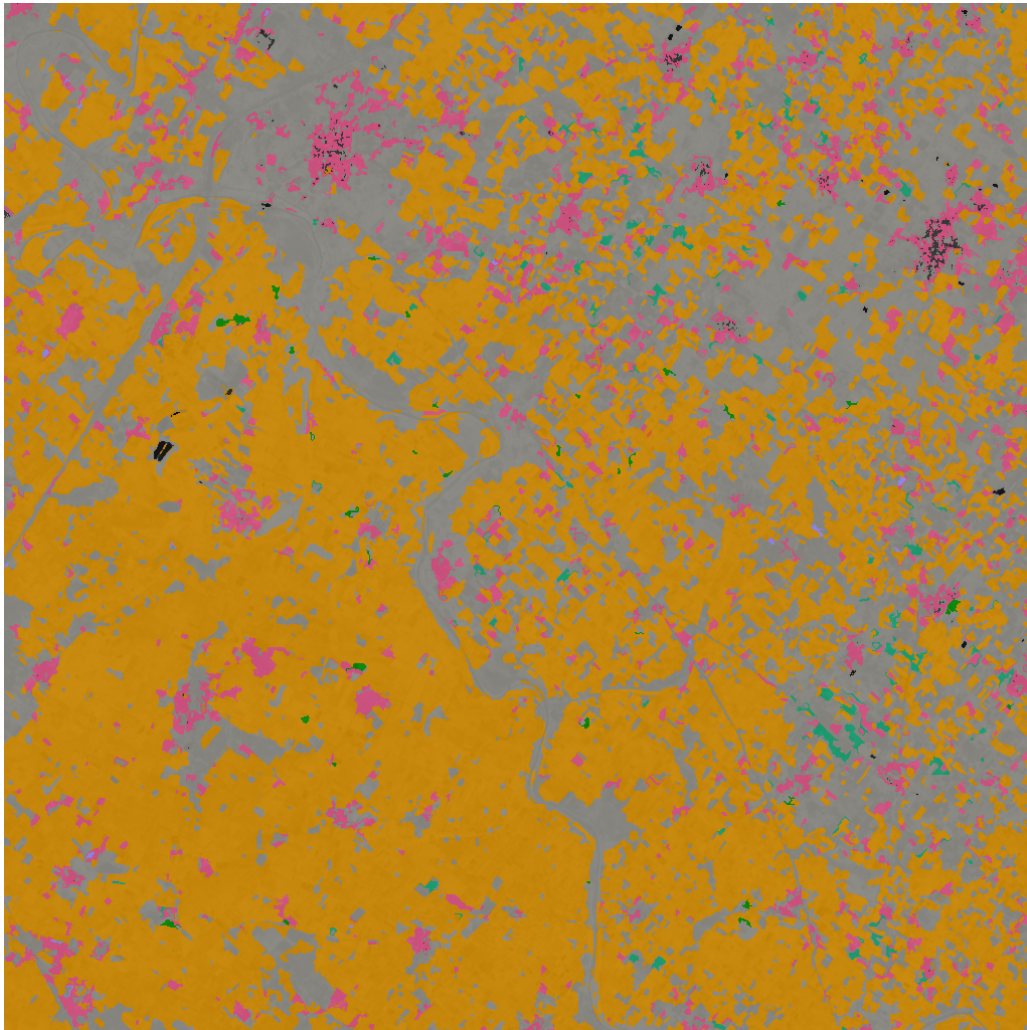


Figure 2 - Field-level crop-type map for a 9x8 km AOI in Uttar Pradesh (rabi 2022-23): wheat (amber), mustard (magenta), sugarcane (teal), maize (green) over the March NDVI backdrop.

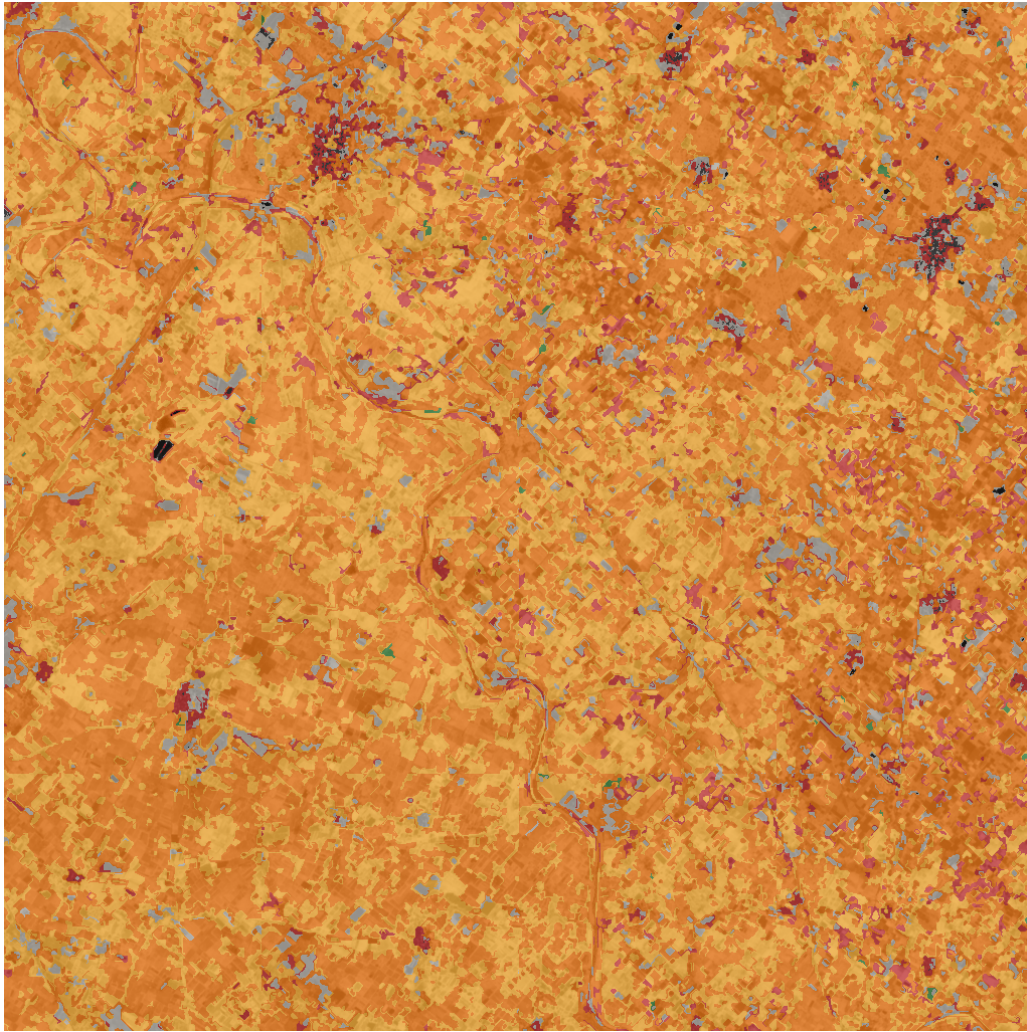


Figure 3 - Per-field irrigation advisory for one dekad (translucent overlay): yellow = watch, orange = irrigation advised, red = severe, grey = out of season.

7 Key Engineering Lessons

- **Chip-level splits or nothing** - pixel- or field-level splits inflate accuracy through spatial autocorrelation; the honest CV number (0.73) cost several “improvements” that evaporated under proper validation.
- **Raw bands beat clever indices** - after 20+ engineered-feature experiments (GLCM texture, phenology metrics, flowering indices, SAR entropy, ...), the only material win was feeding raw band-month values alongside the indices.
- **Physics + ML beats either alone** - the FAO-56 balance provides an interpretable, defensible advisory; the LSTM extends it to weather-sparse operation; fixed fusion rules keep the farmer-facing output auditable.
- **Latency lives in the slowest composite** - GEE computePixels wall-time is bounded by the slowest of ~40 parallel requests (~33 s), not by AOI area, until the per-request payload cap forces tiling.

8 Future Work

- Cross-job composite cache keyed (bbox, month, product).
- Field-delineation model to replace pseudo-field segmentation; finer weather (IMD 0.25 deg / ERA5-Land 9 km); canal command-area rollups of per-field volumes.
- Live mid-season operation (all steps already causal) and a second validated region.

- Breaking the 0.73 OA wall: new labeled wheat/mustard data or hyperspectral (EnMAP) for a recent season.

9 References

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Stack: Python · PyTorch · LightGBM · Google Earth Engine · FastAPI · React · MapLibre GL · rasterio

Website: <https://harshul-pfs.tailde7c85.ts.net/>

GitHub repo: <https://github.com/harshulblr-sys/croptype-krishidrishti-python>